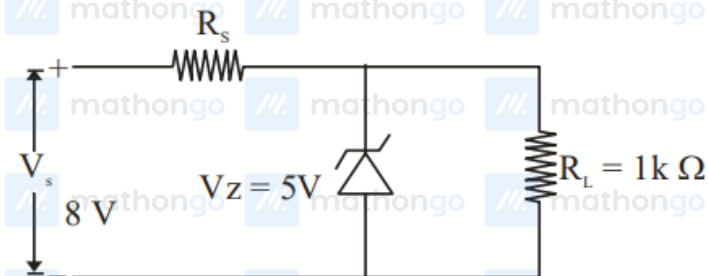


Questions with Answer Keys

MathonGo

Q1 - 2024 (01 Feb Shift 1)

In the given circuit if the power rating of Zener diode is 10 mW, the value of series resistance R_s to regulate the input unregulated supply is:



[We changed options. In official NTA paper no option was correct.]

- (1) $5k\Omega$
- (2) 10Ω
- (3) $1k\Omega$
- (4) None of these

Q2 - 2024 (01 Feb Shift 2)

Conductivity of a photodiode starts changing only if the wavelength of incident light is less than 660 nm. The band gap of photodiode is found to be $\left(\frac{X}{8}\right)$ eV. The value of X is ;

(Given, $h = 6.6 \times 10^{-34}$ Js, $e = 1.6 \times 10^{-19}$ C)

- (1) 15
- (2) 11
- (3) 13
- (4) 21

Q3 - 2024 (27 Jan Shift 1)

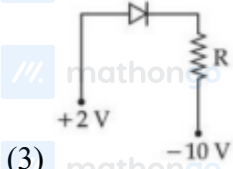
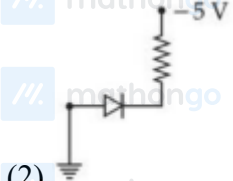
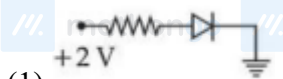
Which of the following circuits is reverse - biased?

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Questions with Answer Keys

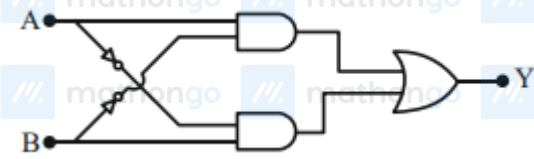
MathonGo



(4) $+2\vec{V} \rightarrow \text{Mu}^{+4}\text{V}$

Q4 - 2024 (27 Jan Shift 2)

The truth table of the given circuit diagram is :



A	B	Y
0	0	1
(1) 0	1	0
1	0	0
1	1	1

A	B	Y
0	0	0
(2) 0	1	1
1	0	1
1	1	0

A	B	Y
0	0	0
(3) 0	1	0
1	0	0
1	1	1

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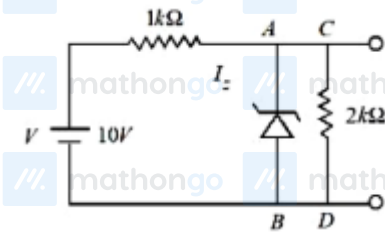
Questions with Answer Keys

MathonGo

///	A	B	Y	///	mathongo	///	mathongo	///	mathongo	///	mathongo	///	mathongo
	0	0	1										
(4)	0	1	1	///	mathongo	///	mathongo	///	mathongo	///	mathongo	///	mathongo
	1	0	1										
///	1	1	0	///	mathongo	///	mathongo	///	mathongo	///	mathongo	///	mathongo

Q5 - 2024 (29 Jan Shift 1)

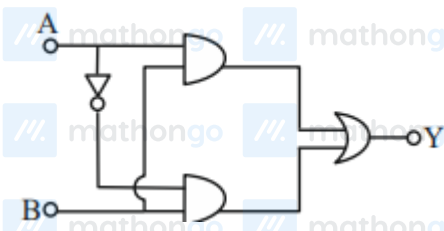
In the given circuit, the breakdown voltage of the Zener diode is 3.0 V. What is the value of I_2 ?



- (1) 3.3 mA
- (2) 5.5 mA
- (3) 10 mA
- (4) 7 mA

Q6 - 2024 (29 Jan Shift 2)

The truth table for this given circuit is :



	A	B	Y
	0	0	1
(1)	0	1	1
	1	0	1
	1	1	0

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Questions with Answer Keys

MathonGo

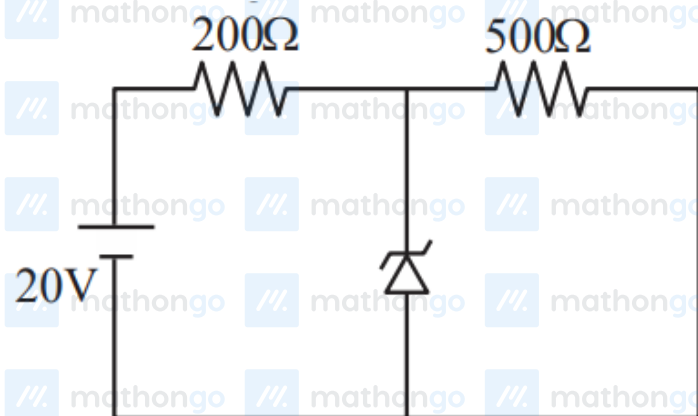
	A	B	Y
	0	0	0
(2)	0	1	1
	1	0	0
	1	1	1

	A	B	Y
	0	0	0
(3)	0	1	0
	1	0	0
	1	1	1

	A	B	Y
	0	0	1
(4)	0	1	0
	1	0	1
	1	1	0

Q7 - 2024 (30 Jan Shift 1)

A Zener diode of breakdown voltage 10 V is used as a voltage regulator as shown in the figure. The current through the Zener diode is



(1) 50 mA

(2) 0

(3) 30 mA

(4) 20 mA

Q8 - 2024 (30 Jan Shift 2)

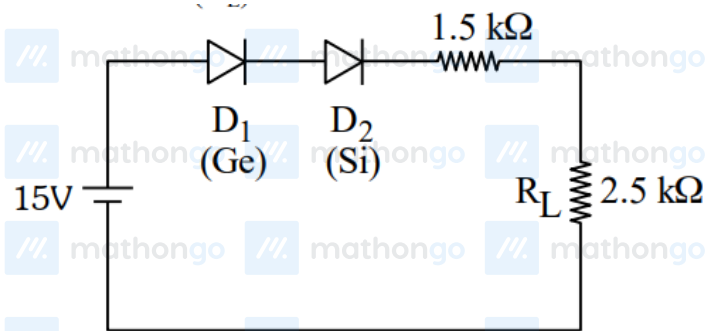
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Questions with Answer Keys

MathonGo

In the given circuit, the voltage across load resistance (R_L) is:



(1) 8.75 V

(2) 9.00 V

(3) 8.50 V

(4) 14.00 V

Q9 - 2024 (31 Jan Shift 1)

Identify the logic operation performed by the given circuit.



(1) NAND

(2) NOR

(3) OR

(4) AND

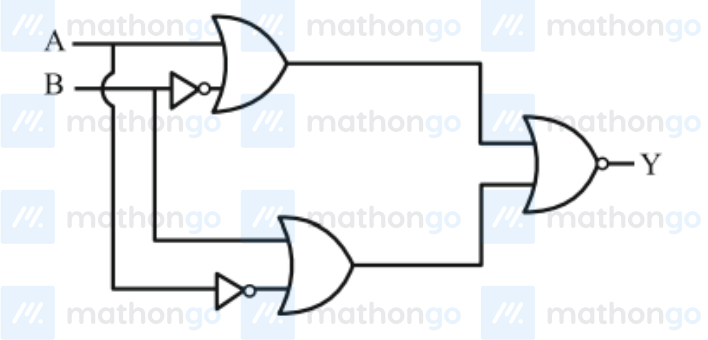
Q10 - 2024 (31 Jan Shift 2)

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Questions with Answer Keys

MathonGo



The output of the given circuit diagram is

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	1

A	B	Y
0	0	0
1	0	1
0	1	1
1	1	0

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	0

A	B	Y
0	0	0
1	0	0
0	1	1
1	1	0

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Answer Key

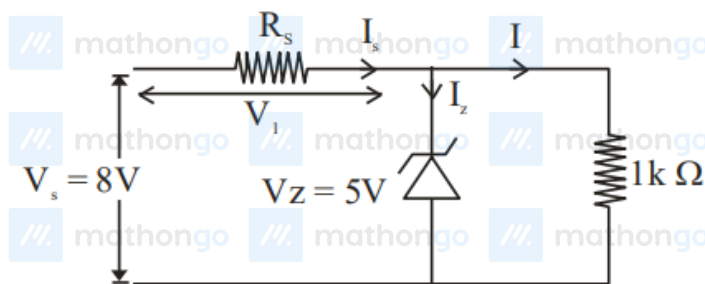
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Solutions

MathonGo

Q1

Pd across R_s

$$V_1 = 8 - 5 = 3 \text{ V}$$

Current through the load resistor

$$I = \frac{5}{1 \times 10^3} = 5 \text{ mA}$$

Maximum current through Zener diode

$$I_{Z \text{ max.}} = \frac{10}{5} = 2 \text{ mA}$$

And minimum current through Zener diode

$$I_{Z \text{ min.}} = 0$$

$$\therefore I_{s \text{ max.}} = 5 + 2 = 7 \text{ mA}$$

$$\text{And } R_{s \text{ min}} = \frac{V_1}{I_{s \text{ max.}}} = \frac{3}{7} \text{ k}\Omega$$

Similarly

$$I_{s \text{ min.}} = 5 \text{ mA}$$

$$\text{And } R_{s \text{ max.}} = \frac{V_1}{I_{s \text{ min.}}} = \frac{3}{5} \text{ k}\Omega$$

$$\therefore \frac{3}{7} \text{ k}\Omega < R_s < \frac{3}{5} \text{ k}\Omega$$

Q2

$$E_g = \frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{660 \times 10^{-9}} \text{ J}$$

$$= \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{660 \times 10^{-9} \times 1.6 \times 10^{-19}} \text{ eV}$$

$$= \frac{15}{8} \text{ eV}$$

So $x = 15$

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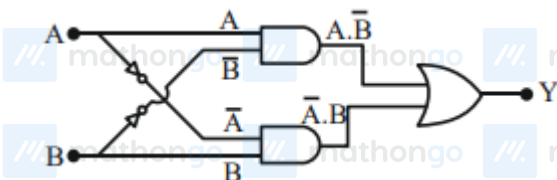
Solutions

MathonGo

Q3

P end should be at higher potential for forward biasing.

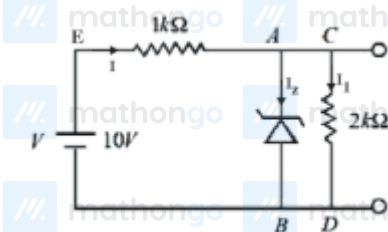
Q4



$$Y = A \cdot \bar{B} + \bar{A} \cdot B$$

This is XOR GATE

Q5



$$V_z = 3 \text{ V}$$

Let potential at $B = 0 \text{ V}$

Potential at E (V_E) = 10 V

$$V_c = V_A = 3 \text{ V}$$

$$I_z + I_1 = I$$

$$I = \frac{10 - 3}{1000} = \frac{7}{1000} \text{ A}$$

$$I_1 = \frac{3}{2000} \text{ A}$$

$$\text{Therefore } I_z = \frac{7 - 1.5}{1000} = 5.5 \text{ mA}$$

Q6

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Solutions

MathonGo



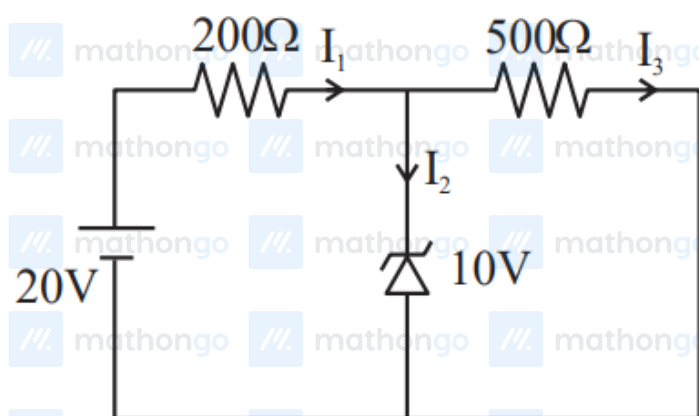
$$Y = A \cdot B + \bar{A} \cdot B$$

$$= (A + \bar{A}) \cdot B$$

$$Y = 1 \cdot B$$

$$Y = B$$

Q7



Zener is in breakdown region.

$$I_3 = \frac{10}{500} = \frac{1}{50}$$

$$I_1 = \frac{10}{200} = \frac{1}{20}$$

$$I_2 = I_1 - I_3$$

$$I_2 = \left(\frac{1}{20} - \frac{1}{50} \right) = \left(\frac{3}{100} \right) = 30 \text{ mA}$$

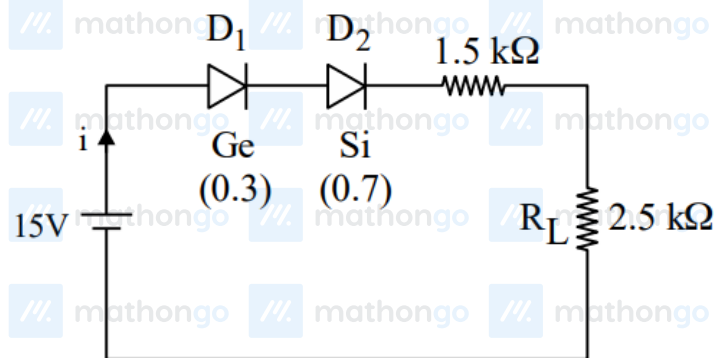
Q8

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Solutions

MathonGo



$$i = \frac{14}{4} = 3.5 \text{ mA}$$

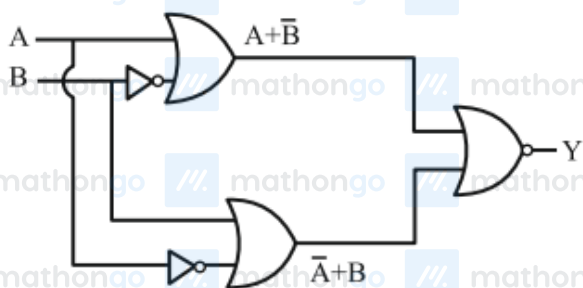
$$V_L = iR_L = 3.5 \times 2.5 \text{ volt} \\ = 8.75 \text{ volt}$$

Q9

$$Y = \overline{\overline{A} \cdot \overline{B}} = \overline{\overline{A}} + \overline{\overline{B}} = A + B$$

(De-Morgan's law)

Q10



$$\text{If } A = 0; \overline{A} = 1$$

$$A = 1; \overline{A} = 0$$

$$B = 0; \overline{B} = 1$$

$$B = 1; \overline{B} = 0$$

$$Y = (A + \overline{B}) + (\overline{A} + B) = \overline{(1 + 1)} = 0$$

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